

Mitigating Texture Bias: A Remote Sensing Super-Resolution Method Focusing on High-Frequency Texture Reconstruction

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Abstract—Super resolution (SR) is an ill-posed problem because one low-resolution image can correspond to multiple high-resolution images. High-frequency details are significantly lost in low-resolution images. Existing deep learning-based SR models excel in reconstructing low frequency and regular textures but often fail to achieve high-quality reconstruction of SR high-frequency textures. These models exhibit bias toward different texture regions, leading to imbalanced reconstruction across various areas. To address this issue and reduce model bias toward diverse texture patterns, we propose a frequency-aware SR method that improves the reconstruction of high-frequency textures by incorporating local data distributions. First, we introduce the frequency-aware transformer (FAT), which enhances the capability of transformer-based models to extract frequency domain and global features from remote sensing images. Moreover, we design a local extremum and variance-based loss function, which guides the model to reconstruct more realistic texture details by focusing on local data distribution. Finally, we construct a high-quality remote sensing SR dataset named RSSR25. We also discover that denoising algorithms can serve as an effective enhancement method for existing public datasets to improve model performance. Extensive experiments on multiple datasets demonstrate that the proposed FAT achieves superior perceptual quality while maintaining high-distortion metrics scores compared with state-of-the-art algorithms. The source code and dataset will be publicly available at: <https://github.com/fengyanzi/FAT>.

Index Terms—Fourier transform, remote sensing images, super resolution (SR), transformer.

I. INTRODUCTION

IN RECENT decades, the rapid advancement of remote sensing technology has significantly propelled the progress of Earth's observation, providing valuable data support for critical fields, such as agricultural monitoring, environmental protection, and urban planning [1], [2], [3], [4]. With the increasing demands of remote sensing applications, the need for high-resolution remote sensing images has concurrently

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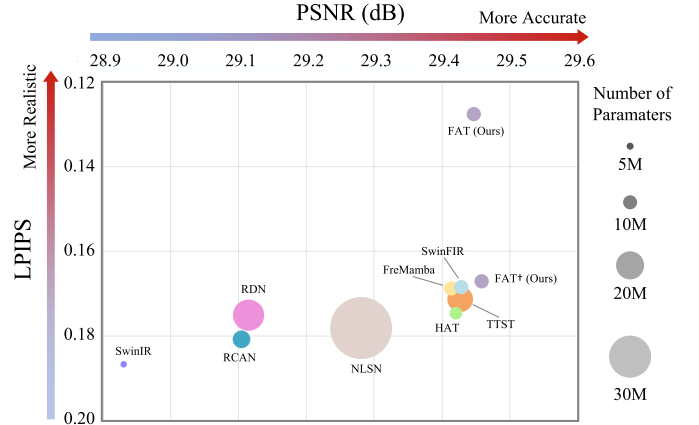


Fig. 1. Comparison of model parameters, PSNR, and LPIPS for various SR methods on the constructed RSSR25 dataset. Higher PSNR indicates greater accuracy, while higher LPIPS reflects perceptual quality closer to reality. Notably, FAT[†] was optimized solely with L_1 loss, where FAT utilized the proposed loss function for improved performance.

grown [5]. However, due to hardware limitations and cost issues, improving the physical resolution of remote sensing images presents significant challenges. To address this issue, remote sensing image super-resolution (SR) technology has emerged, aiming to enhance the visual resolution of images through software methods.

In terms of method design, early efforts primarily focused on achieving image SR through the use of prior knowledge [6]. However, with the advancement of deep learning, methods based on convolutional neural network (CNN), generative adversarial network (GAN), and transformer have demonstrated superior performance compared with traditional approaches [7], [8], [9].

Benefiting from the densely connected residual architecture, CNN-based models demonstrate remarkable capabilities in capturing local spatial features and accurately reconstructing low-frequency textures within remote sensing imagery [10]. However, their limited ability to model global information constrains their overall performance [11], [12]. GANs enable more realistic reconstruction of texture details through adversarial training but may face challenges, such as training instability and mode collapse [13], [14].

Transformer-based methods have attracted attention due to their powerful long-range representation capabilities [15]. Models like Vision Transformer [16] and Swin Transformer [17] have demonstrated outstanding performance in

advanced computer vision tasks, such as object recognition and segmentation. For low-level vision tasks like image SR, transformer-based approaches achieve theoretically global receptive fields by computing correlations across all input tokens via self-attention mechanisms, thereby demonstrating superior performance [18]. However, several challenges remain when applying transformer models to remote sensing image SR. First, a substantial amount of high-frequency details are inevitably lost during the image degradation process [19]. To achieve accurate reconstruction, the model must infer high-frequency information from the remaining low-frequency information. Current models often demonstrate limited capability in extracting frequency-domain features and capturing interfrequency correlations. Second, the full potential of transformer models' global modeling capability has yet to be fully activated, with their effective receptive field remaining constrained [20].

To address these issues, we extend the transformer-based SR model from a frequency learning perspective. The Fourier transform enables comprehensive representation of frequency-domain information across the entire spatial domain, while sparse self-attention mechanisms facilitate efficient global information interaction with reduced computational cost and noise suppression. By integrating these two components, we develop the frequency-domain sparse self-attention (FSSA), which aims to enhance high-frequency texture reconstruction while maintaining computational efficiency. As shown in Fig. 1, this design expands the model's effective receptive field, thereby further activating the potential of transformer-based models for remote sensing image SR.

In terms of loss functions employed, the majority of models utilize distortion-driven losses such as the L_1 loss function to guide model optimization [9], [11], [12], [21], [22], [23], [24]. The L_1 loss function can effectively guide models to achieve outstanding peak signal-to-noise ratio (PSNR) performance [25]. However, remote sensing image SR is inherently an ill-posed problem, where a single low-resolution image can correspond to multiple high-resolution images [26]. Models trained with distortion-driven losses tend to reconstruct each pixel value as the median of its multiple corresponding values [27]. As shown in Fig. 2, while distortion-driven losses work well for low-frequency textures such as smooth surfaces, they tend to reconstruct over-smoothed textures in regions with high-frequency textures [28]. Some models have adopted perceptual-driven losses to enhance image realism, but they may introduce noise and artifacts [29]. Currently, there is still a lack of loss function that effectively balances between realism and accuracy in reconstructed images.

In terms of dataset preparation, the quality of the dataset directly impacts model performance [30]. Many remote sensing image SR models are trained on datasets, such as DOTA [31], AID [32], NWPU-RESISC45 [33], and DIOR [34], which are designed for tasks like image classification, image segmentation, and object detection. During the experiments, we unexpectedly discovered that these datasets are often affected by degradation factors such as noise and compression artifacts. These degradation factors exacerbate the difficulty of reconstructing high-frequency texture regions, challenging the model's SR capabilities.

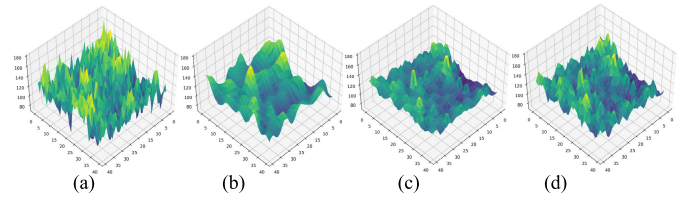


Fig. 2. Distribution diagrams of pixel values for the ground-truth image and the SR results processed by bicubic, HAT, and the proposed FAT, where the x - and y -axes represent pixel coordinates, and the z -axis represents pixel values. Generally, distortion-driven algorithms often fail to restore high-frequency details, resulting in smoother outputs. The proposed method alleviates this issue. (a) HR. (b) Bicubic. (c) HAT. (d) FAT (ours).

To address these challenges, we propose an efficient remote sensing image SR method by focusing on three crucial aspects: model architecture, loss function, and dataset preparation. Our method not only achieves accurate reconstruction of low-frequency textures but also ensures a more realistic rendering of high-frequency textures, thereby reducing the imbalance SR performance across different image regions. The main contributions and innovations of this article are as follows.

- 1) We design the frequency-aware transformer (FAT), which incorporates a U-shaped attention fusion module and multiple transformer layers employing FSSA. These components enhance the model's ability to extract frequency-domain and global features, thereby improving the overall performance of SR models in reconstructing remote sensing image textures.
- 2) We propose the local extremum and variance loss. By supervising the local data distribution of SR images, the proposed loss function helps mitigating the model's bias toward different textures, further enhancing the recovery of high-frequency details.
- 3) A high-quality remote sensing image SR dataset RSSR25 has been constructed, which comprises a training set with 5000 images and a benchmark testing set. In addition, we discover that denoising models can be utilized as a dataset augmentation strategy to reduce the negative impact of noise and compression artifacts on SR models.

The remainder of this article is organized as follows. Section II introduces the related works. Section III details the proposed methods. Section IV discusses the experimental results. Section V concludes this article.

II. RELATED WORKS

This section provides an overview of the existing literature on image SR methods, frequency-domain approaches, the commonly employed loss functions, and the datasets utilized in remote sensing image SR research.

A. SR Methods

CNNs have been extensively studied in the domain of single-image SR. The pioneering work by Dong et al. [35] demonstrated the effectiveness of deep CNNs in mapping low-resolution images to high-resolution ones, significantly improving image restoration quality and computational efficiency over traditional sparse coding techniques. Many studies have combined CNN architectures with residual, dense, and



Fig. 3. Visual comparison of the impact of different loss functions on SR results.

multiscale structures to enhance the model's ability to capture deep features [36], [37], [38], [39], [40], [41]. Some research has further improved feature extraction by incorporating attention mechanisms, such as channel attention [22], [42], self-similar attention [24], and nonlocal sparse attention [23]. To address the complex degradation factors in real-world remote sensing images, Dong et al. [43] and Zhang et al. [44] have enhanced model performance by constructing appropriate degradation models and datasets, respectively. Despite the success in capturing local spatial features and complex representations, CNNs still struggle to resolve the problem of high-frequency detail loss inherent in SR task [8], [45]. To address this issue, researchers have incorporated the GAN mechanism to achieve realistic reconstruction of texture details [46]. Jia et al. [10] and Li et al. [47] utilized attention mechanisms to enhance the ability of GANs to capture complex image details. Considering the interference of noise in real-world image SR, Wang et al. [48] introduced a high-order degradation modeling process to better simulate complex real-world degradations. Meanwhile, Jiang et al. [49] improved the model's performance on noise-affected remote sensing images through the use of mask processing. However, GANs often suffer from unstable training and tend to generate fake textures, which restricts their extensive application in remote sensing images [50].

Transformer has gained attention from the computer vision community due to its ability to capture long-range dependencies through self-attention mechanisms [51]. The Vision Transformer by Dosovitskiy et al. [16] marked the initial application of transformers in vision tasks. Following this, Liu et al. [17] introduced the Swin Transformer, which adapted the transformer architecture to vision tasks with a hierarchical representation strategy and shifted windows, significantly improving its efficiency and effectiveness. Further innovations in transformer-based models have been made to address the specific requirements of low-level vision tasks. For example, Liang et al. [11] developed a benchmark SR model based on the transformer. Lei et al. [9] developed a multistage enhanced transformer to make full use of high-dimensional and low-dimensional features in remote sensing images. Chen et al. [20] extended the performance upper bound of transformer by activating more input pixels. Xiao et al. [52] introduced a top- k selection mechanism to make the long-range dependency modeling more effective. These transformer-based models have demon-

strated remarkable capabilities in feature extraction. However, further research is required to further enhance the capability of long-range modeling and achieve authentic and accurate reconstruction of high-frequency textures.

B. Fourier Transform

Fourier transform can convert images from the spatial domain to the frequency domain, making it suitable for analyzing global frequency information. As a result, the Fourier transform has been applied in fields, such as semantic segmentation [53], image classification [54], and object detection [55]. In low-level vision tasks, the previous work has primarily focused on using the Fourier transform to separate different frequencies. Fritsche et al. [56] achieved more effective real-world image SR by separating high-frequency noise, while Xie et al. [19] introduced a low-complexity branch for low-frequency features to make the SR network more lightweight. Subsequent work further explored Fourier transform's ability to represent global information. Zhang et al. [15] improved model performance by introducing fast Fourier convolution, while Yao et al. [57] enhanced the effective receptive field of the model using fast Fourier convolution. Xiao et al. [58] combined the Fourier transform with the recently popular state space model to reduce the computational cost of extracting frequency-domain features. However, these methods overlook the relationships between different frequency components, leading to high computational costs and the amplification of harmful frequencies. In this study, we introduce sparse attention in the Fourier transform to extract high-frequency features and capture the relationships between different frequency components, while maintaining computational efficiency and suppressing interference from harmful frequencies.

C. Loss Function

The design of loss functions is critical for the performance of SR models. An appropriate loss function can effectively drive the model to achieve both accurate and realistic texture reconstruction. Many benchmark models, such as EDSR [21], NLSN [23], and HAT [20], primarily rely on L_1 loss to achieve better quantitative performance. However, L_1 loss tends to produce over-smoothed textures in regions with high-frequency details [28]. To address this, other models combine multiple loss functions to mitigate bias and achieve higher quality texture reconstruction. For instance, SwinIR [11] employs perceptual loss [59] to improve the visual quality of real-world image SR, while SwinFIR [15] incorporates Charbonnier loss [60] to generate more realistic results.

To further improve the reconstruction of edge and texture details, Yan et al. [61] integrated a target gradient field as an image prior in high-resolution image reconstruction models. Ning et al. [27] introduced image uncertainty to guide the model to focus on textured and edge pixels with high uncertainty. While Abrahamyan et al. [62] developed gradient variance loss, which enforces the model to generate high-resolution images with sharper edges.

However, as shown in Fig. 3, we observed that the gradient loss can misguide optimization, not only leading to

TABLE I
DETAILED SUMMARY OF DATASETS COMMONLY USED FOR TRAINING SR MODELS

Dataset	Functionality	Proposal Year	Total Image Numbers	Original Size	Spatial Resolution
UC Merced	Classification	2010	2100	256×256	0.3m
WHU-RS19	Classification	2011	1005	600×600	>0.5m
NWPU-RESISC45	Classification	2017	31500	256×256	0.2 ~ 30m
AID	Classification	2017	10000	600×600	0.5 ~ 8m
DOTA	Detection	2017	2806	800 ~ 4000	>0.5m*
DIOR	Detection	2020	23463	800×800	0.5 ~ 30m

* The DOTA dataset does not provide information of its spatial resolution. Based on the sources of the images and the dimensions of ground targets such as vehicles within the dataset, we estimated that its actual resolution is no greater than 0.5 meters.

sharper edges but also introducing unexpected noise in certain areas. Perceptual loss performs well but leads to the generation of regular speckled artifacts in high-frequency texture regions [29]. There remains a need for a loss function that can guide the model to achieve high-quality, high-frequency texture reconstruction.

D. Remote Sensing Datasets for SR

High-quality datasets are critical for visual tasks in remote sensing. However, to the best of the authors' knowledge, there remains a lack of a widely accepted benchmark dataset specifically for remote sensing image SR tasks. Most existing SR methods have relied on datasets originally intended for other remote sensing tasks, such as object recognition, image segmentation, and classification. Notable examples include the classification datasets UC Merced [63], WHU-RS19 [64], AID [32], and NWPU-RESISC45 [33], as well as object detection datasets like DOTA [31] and DIOR [34]. Table I provides detailed information about these datasets.

Despite the rich geospatial features in these remote sensing datasets, the maximum spatial resolution of these images is no greater than 0.3 m. Most of the images are not derived from original remote sensing or aerial imagery but are instead sourced from online mapping services, such as Google Earth [31], [32]. During the collection of datasets, many images were improperly resized, compressed, and cropped, which results in a large amount of noise and artifacts. Fig. 4 illustrates some typical examples. Although such noise and artifacts may improve the generalization ability of algorithms in high-level visual tasks, they limit model performance in SR tasks. There is an urgent need for a high-quality benchmark dataset in the field of remote sensing image SR.

III. PROPOSED METHODS

This section provides a comprehensive description of the proposed method. The architecture of the proposed model is detailed in Section III-A. The principles and specifics of the proposed loss function are described in Section III-B. Section III-C offers the information of the proposed dataset and the enhanced methods applied to existing datasets.

A. Overall Structure

Fig. 5 illustrates the overall architecture of the proposed model. The FAT model is composed of three main components. In the shallow feature extraction stage, the



Fig. 4. Examples of images affected by noise and compression artifacts.

model employs a series of 3×3 convolutional layers to extract shallow features $F_{\text{shallow}} \in \mathbb{R}^{H \times W \times C}$ from the input low-resolution image $I_{\text{LR}} \in \mathbb{R}^{H \times W \times 3}$, gradually increasing the number of channels to enrich feature representation. To further enhance the efficiency of feature extraction, we introduce the U-shaped attention fusion block (UAFB), which utilizes a U-shaped mixed attention mechanism for multiscale feature extraction. In the deep feature extraction stage, the model integrates multiple sequential residual transformer blocks (SRTBs). Each SRTB comprises several Swin Transformer layers (STLs) and multiattention transformer (MAT) layers, enhancing the model's ability to extract frequency-domain information and activating the transformer's long-range dependencies. In the final reconstruction stage, the model employs subpixel upsampling to achieve high-resolution reconstruction. The details of each stage are elaborated below.

1) *U-Shaped Attention Fusion Block*: Several advanced high-level vision models utilize CNN–transformer hybrid architectures to enhance model performance and efficiency [65]. These models typically employ convolutional blocks in the initial stages to improve the efficiency of

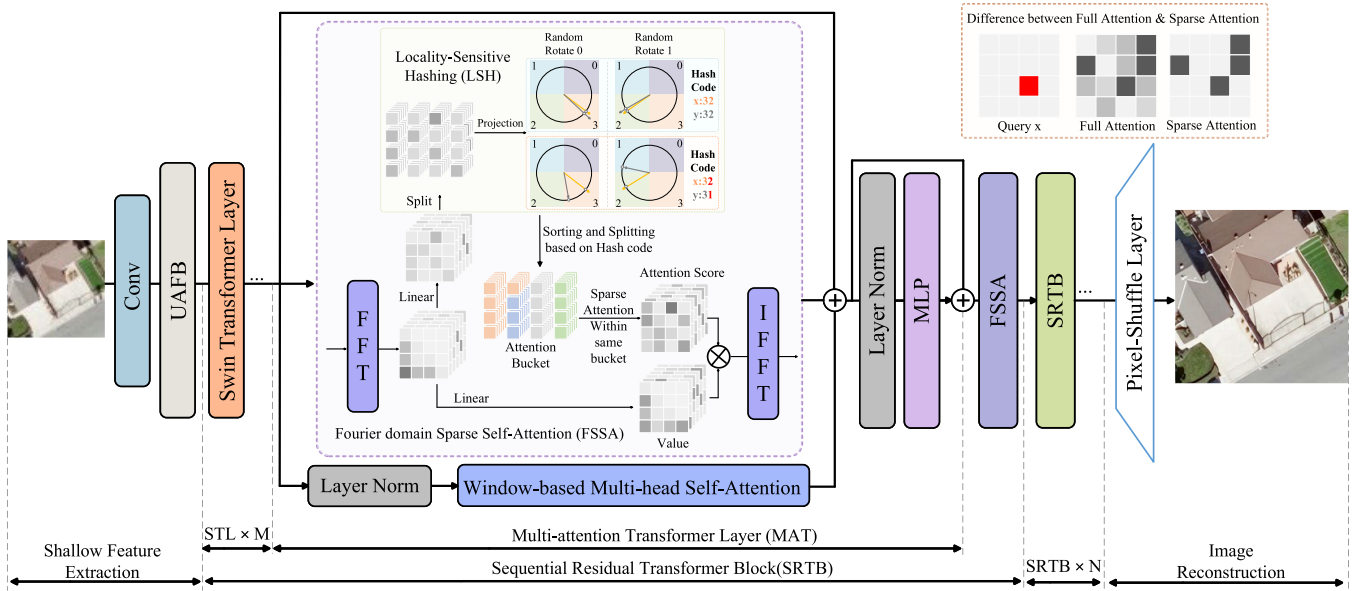


Fig. 5. Main Structure of FAT: 1) Shallow Feature Extraction: Applies 3×3 convolutions for initial feature extraction and integrates a UAFB for efficient multiscale feature representation. 2) Deep Feature Extraction: Utilizes SRTBs with Fourier domain sparse self-attention to extract informative features in the frequency domain. 3) Image Reconstruction: Employs upsampling methods to generate HR outputs.

shallow feature extraction [66], [67]. In addition, some models incorporate U-Net architectures to enhance the extraction of global features [68], [69]. U-Net can be regarded as a natural filter that separates features of different frequencies and scales [70]. Inspired by these approaches, we propose the UAFB to strengthen the model's ability to extract multifrequency features in the shallow feature extraction stage.

During the shallow feature extraction stage, many SR models only apply a 3×3 convolution to the input low-resolution image to increase the number of channels [11], [15], [20]. In contrast, our approach employs UAFB after the 3×3 convolution to extract multiscale features. As shown in Fig. 6, the UAFB module downsamples the input to obtain feature maps at different scales and then applies spatial attention to these feature maps. These features are subsequently fused through channel attention. For the input feature F_{in} , the entire process of this module can be expressed by the following equations:

$$F_M = \text{Upsample}(\text{Gelu}(\text{SA}(\text{Pool}(F_{in})))) \quad (1)$$

$$F_N = \text{Gelu}(\text{SA}(F_{in})) \quad (2)$$

$$F_{out} = \text{Conv}(\text{CA}(\text{Concat}(F_M, F_N))) + F_{in} \quad (3)$$

where F_M and F_N represent the intermediate feature and F_{out} denotes the output of UAFB. CA and SA refer to the channel attention and the spatial attention, respectively. Gelu represents the GELU activation function. Pool and Upsample stand for max pooling and deconvolution. The Conv layer reduces the channel dimensions by half, ensuring that the input and output feature maps of UAFB maintain the same size.

2) *Fourier Domain Sparse Self-Attention*: To capture the relationships among features with different frequencies, we propose the Fourier domain sparse self-attention module. This module leverages the capability of Fourier transform to extract global frequency information while incorporating the efficiency of sparse algorithms.

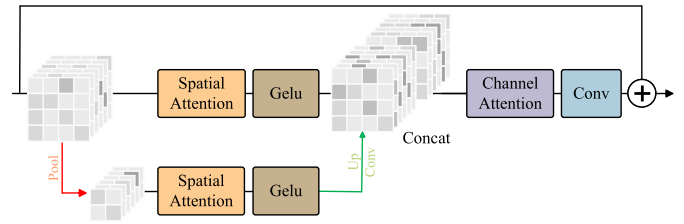


Fig. 6. Main structure of UAFB.

The FSSA module begins by applying a Fourier transform to the input feature map, mapping it from the spatial domain to frequency domain. In the frequency domain, high-frequency information is prominently highlighted. To enhance the efficiency of high-frequency feature extraction and model the relationships among different features, we employ a sparse self-attention mechanism. Traditional self-attention methods compute attention scores by evaluating pairwise interactions among all pixels, leading to high computational costs. However, attention scores are often influenced by a small subset of features most relevant to the query signal [23], [71]. By leveraging sparse techniques to focus on these key features, we not only improve efficiency but also suppress noise effectively.

As shown in Fig. 5, we achieve sparsity in the self-attention mechanism by using angular locality-sensitive hashing (LSH). High-dimensional features are projected onto a hypersphere through random rotations and hash functions, generating hash codes based on their positions. Features are then assigned to different hash buckets according to hash codes. Then, features within each hash bucket are sorted and redistributed into equal-sized attention buckets. The self-attention is applied exclusively within these buckets, significantly reducing global computational complexity while preserving critical feature modeling.

Occasionally, distant features may be incorrectly grouped into the same bucket due to the randomness of

single-projection hashing. To address this, FSSA incorporates multiround hashing. While this approach slightly increases computational cost, it effectively reduces the probability of misassignment. In addition, hash bucket sizes are often imbalanced, which can impair modeling efficiency. To mitigate this issue, features within hash buckets are sorted and redistributed into equal-sized attention buckets. To further ensure effective modeling, cross-bucket attention is allowed between adjacent buckets, preventing relevant features from being excluded due to bucket boundaries.

Finally, values are weighted by the global attention scores, and the enhanced feature map is reconstructed in the spatial domain using an inverse Fourier transform. By appropriately configuring attention bucket sizes, FSSA achieves approximately 93% sparsity in the frequency domain while maintaining effective global self-attention, striking a balance between computational efficiency and performance.

3) *Sequential Residual Transformer Block*: The SRTB constitutes the primary component of the proposed FAT model and largely determines its SR capabilities. To allow for flexible model scaling and to balance efficiency and performance, the deep feature extraction part of the model adopts a two-stage structure. The model includes multiple SRTB modules, each containing several classical STLs and a MAT module [11]. To further enhance the inductive bias of transformer models, an FSSA module is appended at the end. The entire process of SRTB can be expressed as follows:

$$Y = \text{FSSA}(\text{MAT}(\text{STL}^n(X))) \quad (4)$$

where X represents the input feature map and Y denotes the output of SRTB.

The MAT module serves as the core of the FAT model. It efficiently extracts high-frequency information through sparse attention in the frequency domain and enhances long-range attention via an enlarged self-attention window. The MAT module design builds upon the STL by incorporating an FSSA module in parallel with the window-based multihead self-attention (W-MSA) module. The process of the MAT module can be represented as follows:

$$F_M = \text{W-MSA}(\text{LN}(X)) + \text{FSSA}(X) + X \quad (5)$$

$$Y = \text{MLP}(\text{LN}(F_M)) + F_M \quad (6)$$

where F_N and F_M represents the intermediate features, X represents the input feature map, and Y denotes the output of the MAT module. Similar to the standard Swin Transformer block, the MAT employs W-MSA and the multilayer perceptron (MLP).

B. Loss Function

The capability of image SR is highly dependent on the design of the loss function. To further improve the model's SR performance and minimize over-smoothing artifacts, we propose a local extremum and variance loss function. The workflow of the proposed loss function is illustrated in Fig. 7.

1) *Analysis of Loss Function*: Distortion-driven loss functions tend to excel at reconstructing low-frequency textures but face challenges when dealing with high-frequency details

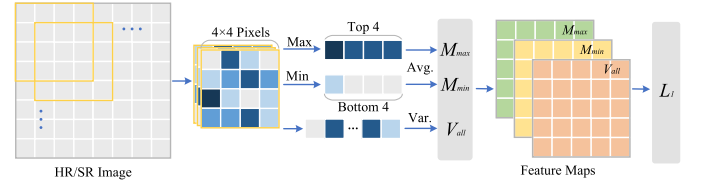


Fig. 7. Workflow of the local extremum and variance loss function is as follows: high-resolution ground-truth images and the SR results of low-resolution images are processed separately to generate local data distribution maps. The local extremum and variance losses are then calculated by computing the mean absolute error between the corresponding local data distribution maps.

in SR tasks. Models trained with the L_1 loss function often produce smoothed reconstructions in high-frequency texture regions, while those trained with the L_2 loss exhibit over-smoothed results [28]. To understand why distortion-driven loss functions demonstrate such bias in reconstructing different textures, we can analyze the problem from the perspective of local data distribution in images.

The SR problem is inherently ill-posed, as a single low-resolution image may correspond to multiple high-resolution images through downsampling. During training, each pixel in the reconstructed SR image could map to multiple plausible values. We can infer that the value of each pixel in the SR image should ideally follow an independent probability distribution p obtained from the training set, with an expected value μ and a variance σ^2 . We assume that the model's pixel predictions follow a distribution $\hat{p}$ with an expected value $\hat{\mu}$ and a variance $\hat{\sigma}^2$. When using the L_1 loss, the model tends to reconstruct each pixel as the median of the possible values [27]. Although the expected value $\hat{\mu}$ aligns with μ , the variance $\hat{\sigma}^2$ significantly decreases, resulting in $\hat{\sigma}^2 \ll \sigma^2$. When L_2 loss is used, the model tends to predict each pixel as the mean μ of the pixel values [27], reducing the variance $\hat{\sigma}^2$ to 0.

Furthermore, for a local patch within the reconstructed SR image (e.g., a 4×4 patch), similar textures in remote sensing images tend to be concentrated within certain regions. For small patches, we can approximate that each pixel follows an independent distribution with varying expectations μ_i but the same variances σ^2 . The variance $\text{Var}(x_{\text{patch}})$ within such a patch is given by

$$\text{Var}(x_{\text{patch}}) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \quad (7)$$

where $x_1, x_2, \dots, x_n$ are the pixel values, n is the number of pixels in the patch, and $\bar{x}$ is the mean pixel value. It is important to note that the variance of the patch pixel values $\text{Var}(x_{\text{patch}})$ is distinct from the variance of the pixel value distribution σ^2 . Assuming that each pixel x_i follows a distribution with expectation μ_i and variance σ^2 , the equation can be expanded as follows:

$$\begin{aligned} \text{Var}(x_{\text{patch}}) &= \frac{1}{n} \sum_{i=1}^n (x_i - \mu_i + \mu_i - \bar{x})^2 \\ &= \frac{1}{n} \sum_{i=1}^n (x_i - \mu_i)^2 + \frac{1}{n} \sum_{i=1}^n (\mu_i - \bar{x})^2 \end{aligned}$$

$$+ \frac{2}{n} \sum_{i=1}^n (x_i - \mu_i)(\mu_i - \bar{x}). \quad (8)$$

Assuming that x_i follows the principle of independent distribution, we have

$$\mathbb{E} \left[\sum_{i=1}^n (x_i - \mu_i)(\mu_i - \bar{x}) \right] = 0. \quad (9)$$

Therefore, the expect of $\text{Var}(x_{\text{patch}})$ can be simplified to

$$\mathbb{E}[\text{Var}(x_{\text{patch}})] = \sigma^2 + \text{Var}(\mu_i) \quad (10)$$

where $\text{Var}(\mu_i)$ reflects the variance of pixel expectations within the patch.

In low-frequency texture regions, the information loss in low-resolution images is minimal, and the corresponding HR images are largely identical. Consequently, the pixel distribution variance σ is inherently small, leading to $\sigma^2 \approx \hat{\sigma}^2 \approx 0$. As a result, the local variance in HR images $\mathbb{E}[\text{Var}^{\text{HR}}(x_{\text{patch}})]$, the local variance of SR results driven by L_1 loss $\mathbb{E}[\text{Var}_{L_1}^{\text{SR}}(x_{\text{patch}})]$, and the local variance of SR results driven by L_2 loss $\mathbb{E}[\text{Var}_{L_2}^{\text{SR}}(x_{\text{patch}})]$ satisfy the relationship

$$\mathbb{E}[\text{Var}^{\text{HR}}(x_{\text{patch}})] \approx \mathbb{E}[\text{Var}_{L_1}^{\text{SR}}(x_{\text{patch}})] \approx \mathbb{E}[\text{Var}_{L_2}^{\text{SR}}(x_{\text{patch}})].$$

Distortion-driven loss functions achieve superior reconstruction of low-frequency textures.

In regions with high-frequency textures, significant high-frequency information is lost in low-resolution images, resulting in significant variations among the corresponding HR images. Specifically, the variance of pixel distribution σ in high-resolution images is relatively large. When optimizing the SR model with L_1 loss, the local pixel distribution variance $\hat{\sigma}^2$ of the SR result is significantly smaller than σ^2 . While L_2 loss drives the distribution variance of pixel values to nearly 0. This leads to the following relationship:

$$\mathbb{E}[\text{Var}^{\text{HR}}(x_{\text{patch}})] \gg \mathbb{E}[\text{Var}_{L_1}^{\text{SR}}(x_{\text{patch}})] > \mathbb{E}[\text{Var}_{L_2}^{\text{SR}}(x_{\text{patch}})].$$

As a result, the expected local variance $\mathbb{E}[\text{Var}^{\text{SR}}(x_{\text{patch}})]$ in SR results is significantly smaller than $\mathbb{E}[\text{Var}^{\text{HR}}(x_{\text{patch}})]$ in ground-truth images. As shown in Fig. 8, under the influence of distortion-driven loss functions, the local variance and local extremum in the SR results differ substantially from those in the ground-truth images. This disparity highlights a mismatch between the local data distributions of the SR results and high-resolution images, which explains the over-smoothing and unrealistic reconstruction of high-frequency textures. An inappropriate loss function design may be the cause of the model's bias toward certain textures.

Given that σ and μ cannot be directly computed, a novel loss function is proposed to guide the model's focus on local variance $\mathbb{E}[\text{Var}(x_{\text{patch}})]$ as an indirect approach to improve σ . Furthermore, incorporating strategies to emphasize local extremum helps the model better capture high-frequency textures. These improvements enhance the data distribution in SR results, reducing the problem of excessive smoothness and improving image quality.

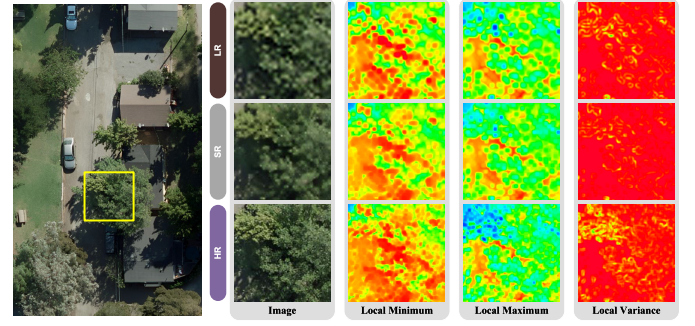


Fig. 8. Local minimum, local maximum, and local variance feature maps of low-resolution, SR (generated using the L_1 loss-driven method HAT), and high-resolution remote sensing images from RSSR25 dataset.

2) *Local Extremum Loss*: First, the high-resolution ground-truth image and the SR image outputted by the model are segmented into two series of localized windows I_w through a sliding window, respectively. For $\times 4$ SR, the window size is set to 4, and the stride is 1. We compute the average value of the four highest pixel values $\text{Max}_4(I_w)$ and the average value of the four lowest pixel values $\text{Min}_4(I_w)$ within all windows in I_w to suppress the influence of noise. These values are then used to construct two local extremum feature maps $\mathcal{M}_{\text{max}}$ and $\mathcal{M}_{\text{min}}$

$$\mathcal{M}_{\text{min}} = \frac{1}{4} \sum \text{Min}_4(I_w) \quad (11)$$

$$\mathcal{M}_{\text{max}} = \frac{1}{4} \sum \text{Max}_4(I_w). \quad (12)$$

Finally, the L_1 loss is utilized to quantify the differences between the local extremum maps of the HR image and those of the SR image. The process can be expressed as follows:

$$\mathcal{L}_{\text{ext-min}} = \mathcal{L}_1(\mathcal{M}_{\text{min}}^{\text{HR}}, \mathcal{M}_{\text{min}}^{\text{SR}}) \quad (13)$$

$$\mathcal{L}_{\text{ext-max}} = \mathcal{L}_1(\mathcal{M}_{\text{max}}^{\text{HR}}, \mathcal{M}_{\text{max}}^{\text{SR}}) \quad (14)$$

$$\mathcal{L}_{\text{ext}} = \mathcal{L}_{\text{ext-min}} + \mathcal{L}_{\text{ext-max}} \quad (15)$$

where $\mathcal{M}_{\text{min}}^{\text{HR}}$ and $\mathcal{M}_{\text{max}}^{\text{HR}}$, and $\mathcal{M}_{\text{min}}^{\text{SR}}$ and $\mathcal{M}_{\text{max}}^{\text{SR}}$ denote the local extremum values for the HR and SR images, respectively.

3) *Local Variance Loss*: Similarly, the image is segmented by employing a sliding window approach with a window size of 4 and a stride of 4. The average variance of all pixels within all windows $\mathcal{V}$ is used to construct local variance feature maps

$$\mathcal{V}^2 = \frac{1}{16} \sum (I_w - \mu)^2 \quad (16)$$

where μ represents the mean value of all pixels in the window.

Finally, the local variance loss is calculated by estimating the L_1 distance between the local variance feature maps derived from the HR image and the SR image. The process can be expressed as follows:

$$\mathcal{L}_{\text{var}} = \mathcal{L}_1(\mathcal{V}^{\text{HR}}, \mathcal{V}^{\text{SR}}) \quad (17)$$

where $\mathcal{V}^{\text{HR}}$ and $\mathcal{V}^{\text{SR}}$ denote the local variance maps for the HR and SR images.



Fig. 9. Sample images of the RSSR25 SR Dataset.

TABLE II
DETAILS OF THE DATASET

Parameters	Details
Data Source	Maxar
Sample Quantity	Train Set: 5000 samples Test Set: 150 samples
High-resolution Image Size	720×720 pixels
Bit Depth	8-bit
Spatial resolution	$0.06 \sim 1.5$ m / pixel
Terrain Involved	Urban areas, Suburbs, Industrial sites, Rivers, Lakes, Deserts

4) *Combined Loss Function*: The final local extremum and variance loss is the sum of the three losses, formulated as follows:

$$\mathcal{L}_{\text{local}} = \lambda_1 \mathcal{L}_{\text{ext}} + \lambda_2 \mathcal{L}_{\text{var}} \quad (18)$$

where λ_1 and λ_2 represent the weights assigned to each component of the loss function.

The impact of these weights will be discussed in Section IV-E. This loss function improves the likelihood of extreme values appearing in the SR images while suppressing output smoothing and distortion, thereby enhancing the model's SR performance across different regions.

C. Dataset

We collected an ultrahigh-resolution remote sensing dataset RSSR25 from multiple sensors. RSSR25 dataset includes a large number of remote sensing images of urban areas, suburbs, industrial sites, and a substantial number of natural landscapes, such as forests, lakes, oceans, and deserts. Specifically, it contains aerial and satellite remote sensing imagery with resolutions ranging from 0.06 to 1.5 m, with more than 70% of the images having a resolution better than 0.15 m. The images were carefully selected and preprocessed to further enhance their quality. The image size was set as 720×720 to facilitate cropping and other processing tasks.

In accordance with classical SR assumptions, we employed bicubic interpolation to downsample high-resolution remote sensing images [72], thereby obtaining the corresponding low-resolution images. Fig. 9 shows some typical samples from the dataset, and the details of the dataset are listed in Table II.

The AID and DOTA remote sensing datasets were not originally intended for SR tasks and suffer from insufficient resolution and various forms of compression noise. However, the rich variety of geospatial targets in these datasets greatly enhances model generalization and improves SR performance across different regions. To make effective use of previous datasets, we applied denoising model SwinIR [11] to process the AID and DOTA datasets, reducing artifacts and noise in the ground-truth images. This denoising step minimizes the impact of noise on the model's ability to reconstruct high-frequency details. The impact of the denoising process on the dataset will be discussed in Section IV-F.

IV. EXPERIMENTS

The performance of the proposed method has been evaluated through a series of experiments. The experimental settings are introduced in Section IV-A, while the comparative results of different methods are presented in Section IV-B. Experimental results under real-world conditions are shown in Section IV-C. Ablation studies in Section IV-D highlight the contributions

and significance of each module in the model. Section IV-E discusses the effectiveness and generalization of the proposed loss function. Finally, Section IV-F explores the impact of dataset augmentation strategies on the edge reconstruction in SR task.

A. Experimental Settings

To enhance the models' generalization and performance, we first pretrained all models on the widely used natural image datasets DIV2K and Flickr2K [72]. Subsequently, further training was conducted using our proposed RSSR25 dataset, along with AID [32] and DOTA v1.0 [31] remote sensing datasets. All images were cropped into 480×480 subimages to better activate the model's ability to capture long-range dependencies. We selected 150 images from the test sets of both RSSR25 and DOTA to evaluate the model's performance. In addition, 300 images were randomly selected across 30 categories from the AID dataset to test the SR capability across different land cover types. The remaining nonoverlapping images from the training datasets were used for training, with a total of 55 111 natural images for pretraining and 56 517 remote sensing images for training. To facilitate model comparison, we used bicubic interpolation to generate low-resolution images from high-resolution natural and remote sensing images. The default downsampling factor was set to 4.

In terms of the hyperparameters of FAT, the final architecture of FAT stacked six SRTB modules, each comprising three STL modules and one MAT module, totaling 24 transformer modules. The number of channels in FAT was set to 128. To ensure a fair comparison, the competing transformer-based models were configured with the same number of transformer modules and channels. In the MAT modules, the window size for self-attention was set to 6. For MAT within each SRTB module, the window size for W-MSA was set to 12, and the number of W-MSA heads was configured to 8, enabling more effective cross-window information interaction. In the FSSA module, random rotation was performed four times for hash projection, and the size of each attention bucket was set to 120, reducing computational complexity through sparsity.

For the training procedure, we ensured a fair comparison by pretraining all models for 200 epochs on natural images, followed by 400 epochs of training on remote sensing images. FAT applied the proposed loss function introduced in Section III-B, constructing a composite loss function that can be mathematically represented as follows:

$$L = L_1 + \alpha L_{\text{perceptual}} + \beta L_{\text{local}} \quad (19)$$

where $L_{\text{perceptual}}$ is the perceptual loss and L_{local} is the local extremum and variance loss. The weight of loss function was configured as $\alpha = 0.001$ and $\beta = 0.1$. The effectiveness of the local extremum and variance loss will be validated in Section IV-E. Given that many comparison models are trained solely using the L_1 loss, to ensure a fair validation of our model's effectiveness, we also conducted tests on the $\text{FAT}^\dagger$ model trained exclusively with the L_1 loss. Models were optimized using the Adam optimizer with parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1e-8$. The learning rate was

set to 1×10^{-4} for pretraining and 5×10^{-5} for subsequent training. All experiments were conducted on a computer equipped with an NVIDIA A100 GPU and two additional computers equipped with NVIDIA RTX 4090 GPUs and Intel i9-14900K CPUs.

B. Evaluation of the Proposed Model

Several metrics were used to evaluate the performance of FAT, including PSNR, structural similarity index (SSIM) [73], and learned perceptual image patch similarity (LPIPS) [74]. PSNR was calculated on the luminance channel Y in the YCbCr color space. These metrics provide a comprehensive evaluation of the image reconstruction quality through quantitative assessments.

To evaluate the effectiveness of the proposed model, we conducted extensive comparisons with various state-of-the-art SR methods. These include CNN-based models, such as EDSR [21], RCAN [22], RDN [36], and NLSN [23], as well as transformer-based models like SwinIR [11], HAT [20], and TTST [52], which was specifically developed for remote sensing. In addition, frequency learning-based SwinFIR [15] and FreMamba [58] were also included in the evaluation.

1) *Quantitative Evaluation:* Table III presents the average PSNR, SSIM, and LPIPS scores across DOTA, AID, and proposed RSSR25 datasets. The model parameters and FLOPs are also provided for comparison. Fig. 1 provides a visual comparison of the results. Our FAT model not only achieves the highest average PSNR and SSIM values but also the lowest LPIPS score. For example, compared with the impressive TTST model, the $\text{FAT}^\dagger$ model demonstrates a 0.032-dB improvement in PSNR and achieves better SSIM and LPIPS performance on the RSSR25 dataset. Moreover, integrating the proposed local extremum and variance losses into the FAT model reduces LPIPS by a remarkable 23.5% with only a slight decrease of 0.016 dB in PSNR performance. It is noteworthy that frequency-domain learning-based SR methods have demonstrated competitive performance. Both SwinFIR and FreMamba outperform the impressive HAT model, highlighting the potential of frequency-domain learning in SR tasks. Our proposed FAT further captures the internal information and correlations within the frequency domain, achieving high-pixel fidelity and perceptual quality.

Moreover, we evaluated the generalization performance of SR models on the AID dataset, which consists of 30 different scene categories. Table IV summarizes the PSNR and SSIM performance for each method across these categories. The results indicate that both the FAT and $\text{FAT}^\dagger$ models outperform other approaches in most categories, delivering the highest PSNR and SSIM scores. Under the supervision of the proposed loss functions, the FAT model effectively focuses on high-frequency features, thereby further improving the SSIM performance. This underscores the model's exceptional capability and generalization in reconstructing high-quality remote sensing images.

We evaluated the performance of the model at different SR scales. In addition to the default $\times 4$ SR experiment, we conducted experiments at $\times 2$ and $\times 3$ SR on the

TABLE III

QUANTITATIVE PSNR/SSIM/LPIPS COMPARISON WITH SOTA SR MODELS ON RSSR25, AID, AND DOTA V1.0 TEST SET. THE BEST AND SECOND BEST PSNR/SSIM METRICS IN EACH CATEGORY ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Type	Methods	#Param. (M)	FLOPs (G)	RSSR25			AID			DOTA		
				PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Baseline	Bicubic	-	-	27.025	0.7332	0.3729	27.673	0.7375	0.4039	26.931	0.9280	0.3575
CNN-based	EDSR	0.779	17.941	28.550	0.7861	0.2064	29.089	0.7918	0.2196	28.890	0.8440	0.1589
	RCAN	12.615	186.896	29.106	0.8020	0.1808	29.637	0.8099	0.1905	29.862	0.8679	0.1286
	RDN	22.271	327.303	29.117	0.8031	0.1752	29.660	0.8110	0.1854	29.890	0.8697	0.1241
	NLSA	44.156	738.972	29.282	0.8081	0.1787	29.758	0.8138	0.1855	30.015	0.8723	0.1239
Mamba-based	FreMamba	8.582	129.944	29.412	0.8126	0.1689	29.869	0.8176	0.1775	30.263	0.8770	0.1180
Transformer-based	SwinIR	4.588	72.805	28.932	0.7988	0.1868	29.462	0.8055	0.1963	29.549	0.8617	0.1354
	HAT	9.019	136.657	29.422	0.8123	0.1747	29.808	0.8167	0.1876	30.172	0.8756	0.1245
	TTST	18.305	270.149	29.428	0.8119	0.1713	29.829	0.8170	0.1812	30.195	0.8760	0.1219
	SwinFIR	7.448	113.078	29.430	0.8132	0.1686	29.896	0.8195	0.1776	30.412	0.8806	0.1151
	FAT $\uparrow$ (Ours)	10.192	153.513	29.460	0.8148	0.1672	29.936	0.8210	0.1749	30.505	0.8811	0.1141
	FAT(Ours)	10.192	153.513	29.444	0.8167	0.1278	29.917	0.8231	0.1335	30.481	0.8836	0.0887

* FAT $\uparrow$ was optimized solely with L_1 loss, where FAT utilized the proposed loss function for improved performance.

TABLE IV

QUANTITATIVE PSNR/SSIM COMPARISON WITH SOTA SR MODELS ON 30 SCENE CATEGORIES OF THE AID TEST SET. THE BEST AND SECOND-BEST PSNR/SSIM METRICS IN EACH CATEGORY ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Category	Bicubic		EDSR		RCAN		RDN		NLSN		SwinIR		HAT		TTST		FreMamba		SwinFIR		FAT (Ours)		FAT $\uparrow$ (Ours)	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Airport	26.299	0.7326	27.714	0.7924	28.279	0.8123	28.335	0.8133	28.456	0.8162	28.079	0.8063	28.450	0.8185	28.482	0.8185	28.531	0.8193	28.546	0.8217	28.562	0.8244	28.580	0.8222
Bareland	33.461	0.7714	34.365	0.7945	34.528	0.7988	34.514	0.7976	34.606	0.8006	34.499	0.7986	34.613	0.8024	34.622	0.8022	34.667	0.8017	34.593	0.8010	34.617	0.8032	34.712	0.8042
Baseball Field	29.069	0.8015	30.666	0.8467	31.146	0.8572	31.174	0.8588	31.288	0.8608	30.951	0.8542	31.334	0.8625	31.376	0.8621	31.372	0.8626	31.500	0.8653	31.390	0.8646	31.467	0.8647
Beach	33.723	0.8518	34.654	0.8695	34.952	0.8730	34.775	0.8722	34.916	0.8737	34.856	0.8735	35.153	0.8767	35.152	0.8764	35.137	0.8768	34.884	0.8759	35.107	0.8771	35.240	0.8780
Bridge	28.558	0.8161	30.810	0.8711	31.464	0.8842	31.442	0.8849	31.511	0.8851	31.211	0.8800	31.645	0.8873	31.678	0.8876	31.743	0.8885	31.754	0.8893	31.815	0.8924	31.795	0.8905
Center	26.175	0.7400	28.248	0.8218	29.030	0.8466	29.057	0.8484	29.162	0.8505	28.762	0.8397	29.259	0.8540	29.226	0.8538	29.213	0.8525	29.380	0.8573	29.417	0.8630	29.331	0.8581
Church	23.570	0.6661	25.138	0.7576	25.822	0.7866	25.811	0.7875	25.938	0.7914	25.567	0.7764	25.951	0.7945	26.014	0.7964	26.017	0.7958	26.117	0.8010	26.110	0.8059	26.094	0.8005
Commercial	24.890	0.6946	26.005	0.7506	26.421	0.7696	26.458	0.7713	26.553	0.7747	26.278	0.7642	26.558	0.7772	26.639	0.7789	26.613	0.7780	26.651	0.7808	26.668	0.7854	26.643	0.7813
Dense Residential	23.577	0.6499	24.629	0.7136	25.070	0.7359	25.079	0.7373	25.202	0.7421	24.895	0.7292	25.194	0.7439	25.273	0.7458	25.271	0.7468	25.334	0.7498	25.318	0.7537	25.314	0.7501
Desert	39.034	0.8981	40.272	0.9127	40.517	0.9161	40.763	0.9164	40.971	0.9179	40.675	0.9170	40.644	0.9178	40.600	0.9189	41.093	0.9195	40.842	0.9169	40.980	0.9193	40.991	0.9198
Farmland	31.697	0.8061	34.239	0.8638	35.568	0.8884	35.511	0.8858	35.721	0.8909	35.053	0.8797	35.785	0.8922	35.825	0.8960	35.822	0.8932	35.800	0.8931	36.025	0.9000	36.006	0.8992
Forest	28.448	0.6353	29.133	0.6564	29.313	0.6713	29.357	0.6695	29.412	0.6745	29.342	0.6722	29.420	0.6791	29.488	0.6807	29.492	0.6813	29.438	0.6788	29.435	0.6878	29.495	0.6847
Industrial	24.435	0.6707	26.305	0.7526	27.089	0.7839	27.062	0.7823	27.267	0.7910	26.866	0.7747	27.268	0.7932	27.286	0.7934	27.338	0.7937	27.445	0.7997	27.410	0.8023	27.440	0.7995
Meadow	34.008	0.7545	35.055	0.7861	35.335	0.7940	35.252	0.7934	35.297	0.7948	35.210	0.7915	35.393	0.7961	35.407	0.7967	35.385	0.7970	35.133	0.7962	35.246	0.7994	35.438	0.8001
Medium Residential	27.126	0.7239	28.433	0.7849	28.867	0.8005	28.886	0.8019	28.977	0.8048	28.752	0.7975	29.048	0.8074	29.088	0.8080	29.050	0.8071	29.088	0.8087	29.114	0.8126	29.135	0.8106
Mountain	29.375	0.7344	30.162	0.7609	30.392	0.7699	30.342	0.7699	30.419	0.7731	30.309	0.7694	30.462	0.7754	30.454	0.7752	30.485	0.7752	30.394	0.7748	30.429	0.7788	30.490	0.7780
Parking	23.533	0.7547	24.487	0.8083	25.193	0.8336	25.153	0.8344	25.441	0.8425	24.912	0.8279	25.485	0.8457	25.594	0.8486	25.699	0.8519	25.860	0.8552	25.983	0.8609	25.868	0.8575
Park	27.964	0.7250	29.008	0.7679	29.395	0.7821	29.426	0.7841	29.435	0.7856	29.297	0.7803	29.486	0.7881	29.509	0.7879	29.513	0.7879	29.520	0.7892	29.548	0.7930	29.560	0.7909
Playground	28.899	0.7987	31.547	0.8770	32.611	0.8991	32.697	0.9008	32.692	0.8990	32.205	0.8899	32.888	0.9042	32.870	0.9045	32.966	0.9064	33.022	0.9072	33.110	0.9105	33.063	0.9091
Pond	30.916	0.8399	32.120	0.8637	32.438	0.8670	32.463	0.8705	32.479	0.8708	32.340	0.8677	32.549	0.8724	32.548	0.8724	32.561	0.8721	32.481	0.8705	32.532	0.8733	32.603	0.8736
Port	22.890	0.7336	23.954	0.7836	24.511	0.8046	24.543	0.8068	24.666	0.8101	24.344	0.7998	24.686	0.8120	24.763	0.8132	24.835	0.8167	24.901	0.8188	24.887	0.8248	24.906	0.8202
Railway Station	25.443	0.6881	27.011	0.7671	27.938	0.8051	28.011	0.8069	28.075	0.8086	27.630	0.7944	28.130	0.8126	28.231	0.8145	28.269	0.8140	28.362	0.8188	28.374	0.8220	28.326	0.8194
Resort	25.238	0.7039	26.426	0.7575	26.965	0.7775	27.021	0.7800	27.099	0.7818	26.798	0.7733	27.147	0.7854	27.218	0.7872	27.123	0.7849	27.258	0.7888	27.173	0.7897	27.176	0.7869
River	28.647	0.7214	29.887	0.7628	30.206	0.7754	30.205	0.7751	30.242	0.7775	30.166	0.7739	30.294	0.7798	30.300	0.7792	30.351	0.7806	30.283	0.7794	30.313	0.7852	30.368	0.7828
School	23.741	0.6840	24.974	0.7492	25.461	0.7696	25.486	0.7719	25.608	0.7758	25.338	0.7672	25.653	0.7794	25.632	0.7795	25.644	0.7795	25.779	0.7839	25.626	0.7843	25.647	0.7818
Sparse Residential	27.606	0.6704	28.814	0.7158	29.172	0.7303	29.206	0.7304	29.274	0.7334	29.094	0.7278	29.288	0.7356	29.293	0.7351	29.311	0.7352	29.321	0.7357	29.285	0.7408	29.358	0.7385
Square	27.994	0.7727	29.510	0.8319	30.027	0.8461	30.066	0.8482	30.133	0.8498	29.862	0.8433	30.185	0.8521	30.194	0.8512	30.252	0.8518	30.232	0.8534	30.277	0.8557	30.285	0.8545
Stadium	22.745	0.6834	24.285	0.7677	25.185	0.8041	25.377	0.8110	25.449	0.8124	24.906	0.7971	25.747	0.8283	25.594	0.8199	25.805	0.8298	26.209	0.8415	26.022	0.8438	26.049	0.8382
Storage Tanks	25.334	0.7241	26.978	0.7925	27.517	0.8101	27.568	0.8115	27.636	0.8132	27.316	0.8046	27.691	0.8158	27.674	0.8151	27.644	0.8145	27.789	0.8175	27.705	0.8197	27.731	0.8178
Viaduct	25.805	0.6775	27.856	0.7724	28.693	0.8049	28.753	0.8077	28.818	0.8103	28.347	0.7942	28.828	0.8122	28.826	0.8119	28.870	0.8127	28.969	0.8162	29.017	0.8203	28.967	0.8174
Average	27.673	0.7375	29.089	0.7918	29.637	0.8099	29.660	0.8110	29.758	0.8138	29.462	0.8055	29.808	0.8167	29.829	0.8170	29.869	0.8176	29.896	0.8195	29.917	0.8231	29.936	0.8210

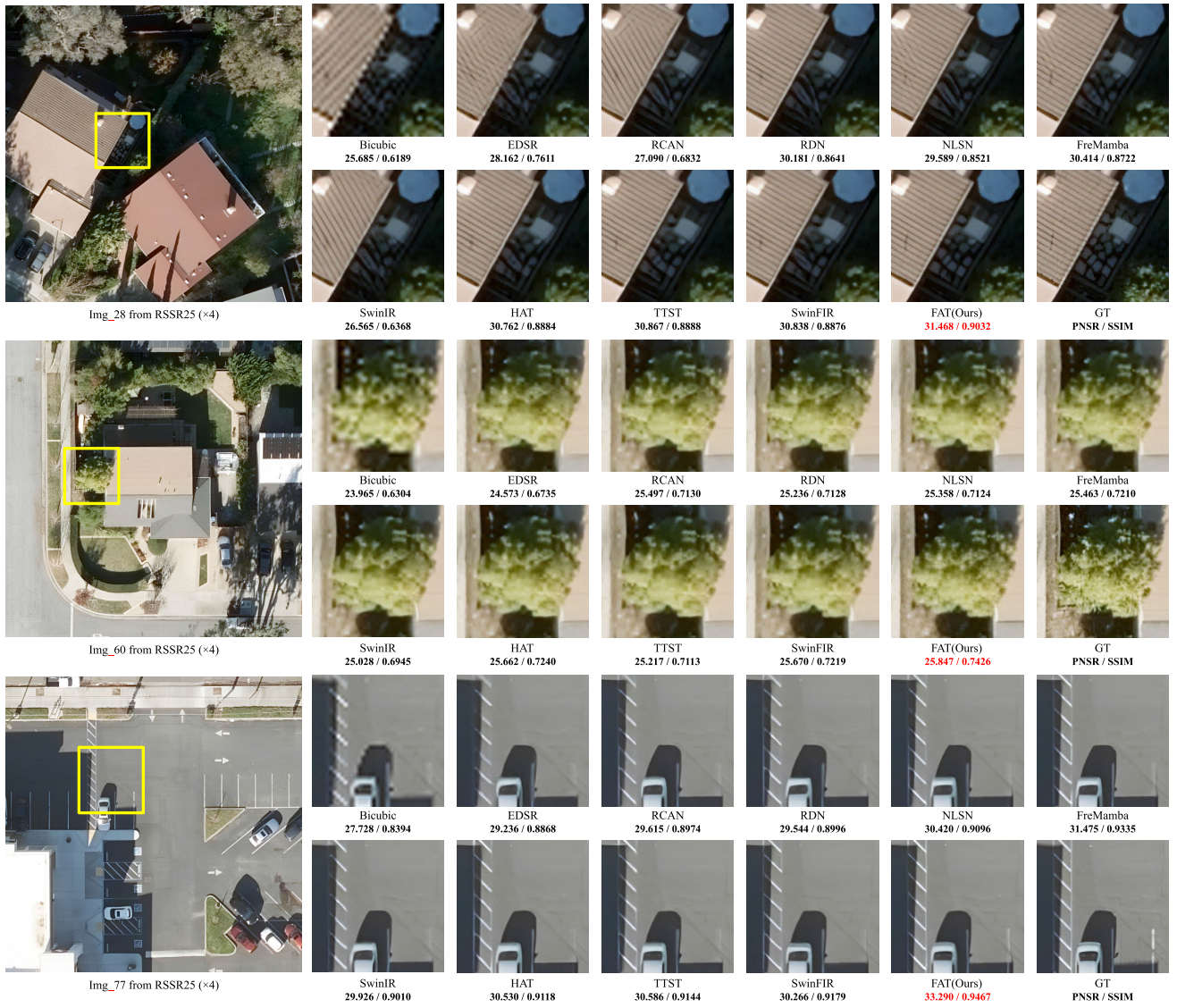


Fig. 10. Visual comparison (x4) on RSSR25 dataset. The best PSNR/SSIM is highlighted in Red. Zoomed-in view for better comparison.

TABLE V
MULTISCALE SR EXPERIMENT ON RSSR25

Methods	×2		×3		×4	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Bicubic	34.253	0.9298	29.056	0.8145	27.025	0.7332
EDSR	36.046	0.9451	31.839	0.8569	28.550	0.7861
RCAN	37.210	0.9497	32.955	0.8647	29.106	0.8020
RDN	37.263	0.9500	32.980	0.8652	29.117	0.8031
NLSA	37.519	0.9532	33.324	0.8684	29.282	0.8081
SwinIR	37.196	0.9486	32.928	0.8641	28.932	0.7988
HAT	37.806	0.9551	33.541	0.8720	29.422	0.8123
TTST	37.869	0.9555	33.559	0.8728	29.428	0.8119
FreMamba	37.910	0.9562	33.570	0.8741	29.412	0.8126
SwinFIR	37.921	0.9563	33.612	0.8745	29.430	0.8132
FAT(Ours)	37.975	0.9570	33.686	0.8752	29.460	0.8148

degradation. In preparing the training data, we adopted the high-order degradation process proposed by Wang et al. [48] instead of bicubic downsampling. For evaluation metrics,

we utilized three widely used no-reference metrics: perceptual index (PI) [75], natural image quality evaluator (NIQE) [76], and average gradient (AG), to comprehensively assess image quality from various perspectives.

Table VI presents the SR performance of various methods on the RSSR25 dataset. The results demonstrate that our FAT model achieves commendable performance across all three metrics. Notably, some CNN-based methods, such as RCAN, which showed moderate performance in simulation experiments, exhibited comparable results to transformer-based methods in real-world scenarios. Specifically, the proposed FAT model outperformed the excellent TTST by achieving a 0.298 reduction in the NIQE score, demonstrating superior generalization capability and adaptability to real-world conditions.

Fig. 13 illustrates a visual comparison of real-world SR results, using a “stop” character marking sprayed on a road surface as an example. While CNN-based approaches struggled with character gap reconstruction, transformer-based methods demonstrated superior performance in rendering straight and

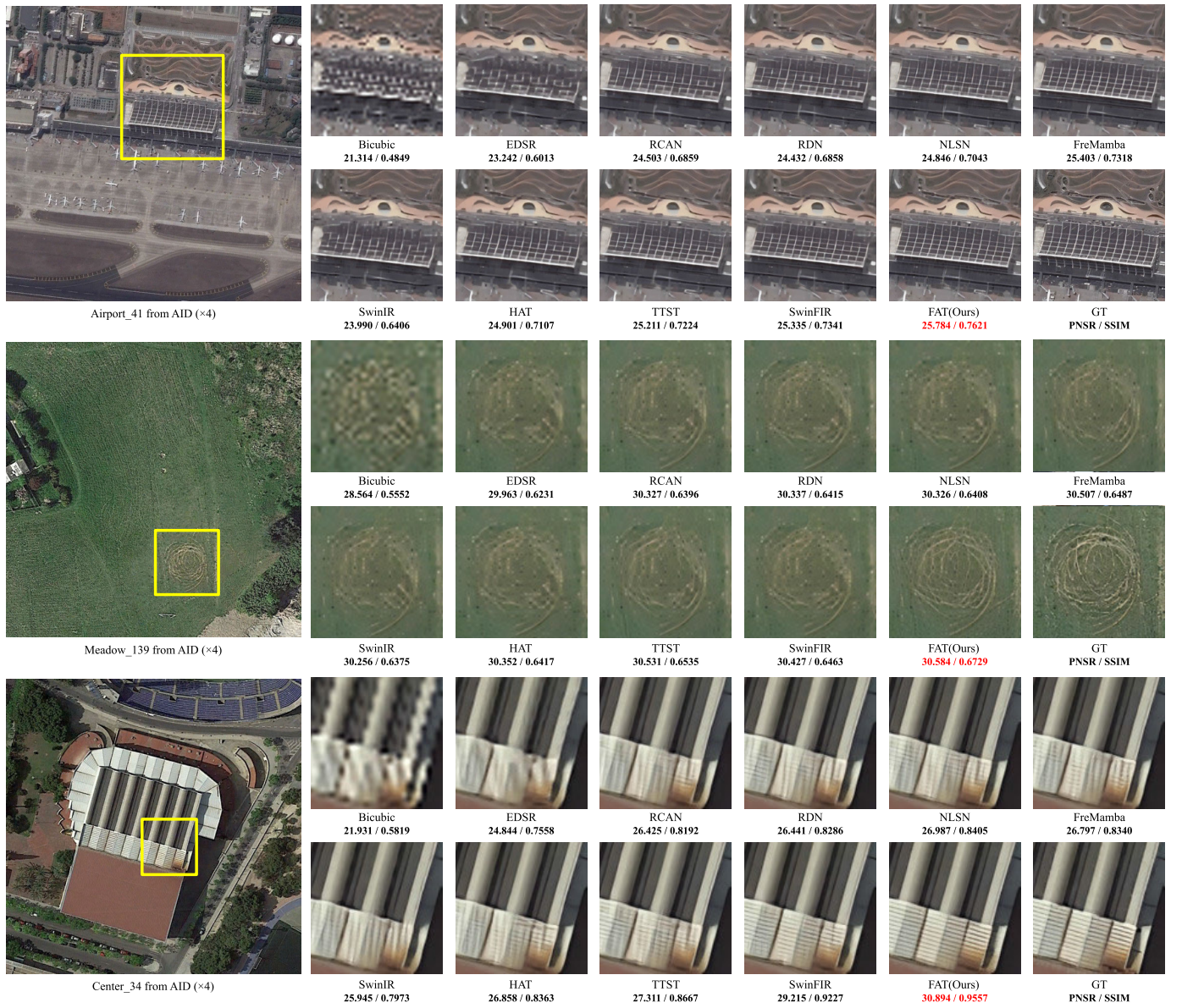


Fig. 11. Visual comparison (x4) on AID dataset. The best PSNR/SSIM is highlighted in Red. Zoomed-in view for better comparison.

smooth character contours through their long-range attention mechanism. However, these transformer methods exhibited limitations in maintaining consistent internal edge reconstruction. The proposed FAT model demonstrates significant advancements through its efficient frequency-domain feature extraction. It successfully achieves dual improvements in both intercharacter gap reconstruction and internal edge restoration, while also effectively recovering intricate character fissures. This comprehensive performance enhancement underscores the model's robustness in handling complex real-world SR tasks.

D. Ablation Study

In the ablation study, we thoroughly evaluated the model design and the parameter configurations of FAT. Notably, all models were trained solely with the L_1 loss function and evaluated on the RSSR25 test set.

TABLE VI
QUANTITATIVE COMPARISON OF REAL-WORLD
SR ON RSSR25

Methods	AG $\uparrow$	PI $\downarrow$	NIQE $\downarrow$
Bicubic	1.594	9.022	10.164
EDSR	1.616	8.331	9.148
RCAN	1.817	8.275	9.085
RDN	1.761	8.208	8.929
NLSA	1.698	8.379	9.117
SwinIR	1.647	8.389	9.118
HAT	1.771	8.260	8.993
TTST	1.742	8.162	8.843
FreMamba	1.804	8.338	9.159
SwinFIR	1.896	8.279	9.168
FAT(Ours)	2.168	7.812	8.545

We conducted ablation tests on three key components of FAT to verify their effectiveness, as shown in Table VII.



Fig. 12. Visual comparison ($\times 4$) on DOTA v1.0 dataset. The best PSNR/SSIM is highlighted in RED. Zoomed-in view for better comparison.

When the UAFB module was removed, the PSNR performance dropped by 0.048 dB. Removing the FSSA attention mechanism at the end of the SRTB module caused a 0.196 dB decrease in PSNR. When the FSSA module parallel to W-MSA in the MAT module was removed, PSNR dropped significantly

by 0.279 dB. The UAFB module naturally functions as a filter, effectively extracting and aggregating features across different scales and frequencies. The FSSA module employs an efficient FSSA mechanism to capture global frequency information and model feature relationships. This design significantly improves

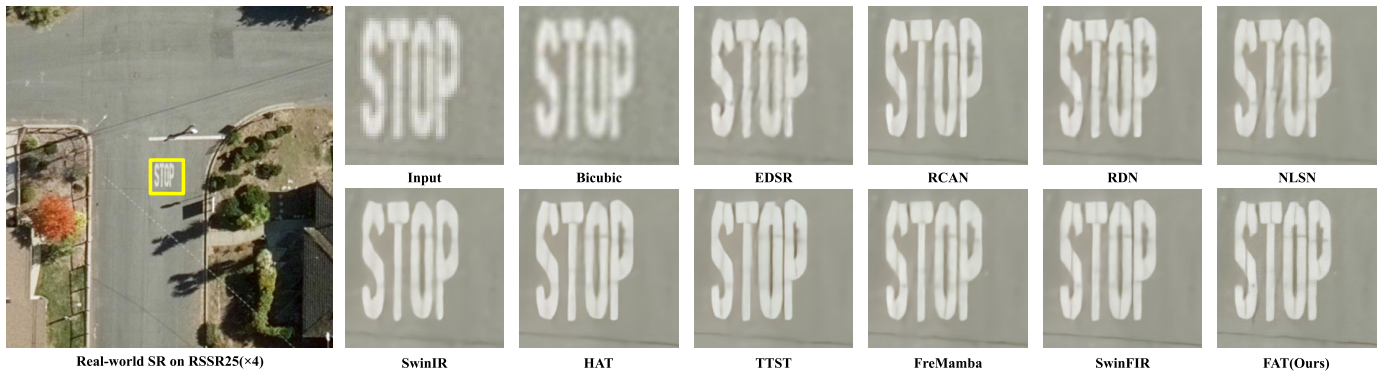


Fig. 13. Real-world SR experiment ($\times 4$) on RSSR25 dataset. Zoomed-in view for better comparison.

interwindow information interaction and extraction efficiency in W-MSA, while further activating the long-range attention mechanism within the transformer architecture. These three components work synergistically to improve the performance of transformer-based models in SR tasks.

To better understand why the proposed FAT network achieves high-quality SR, we employed the local attribution maps (LAMs) [77], which were specifically designed for SR tasks. LAM uses gradient integration to identify input pixels that contribute the most to the reconstruction of selected regions, with deeper red indicating more significant contributions. Intuitively, a broader range and higher efficiency of utilized information lead to better SR performance [77]. As illustrated in Fig. 14, NLSN achieves a larger effective receptive field through nonlocal sparse attention, surpassing many CNN-based methods. Although the transformer-based TTST expands the receptive field to encompass all image pixels, its information utilization efficiency requires further improvement. In contrast, our proposed FAT model activates nearly all pixels in the image with meaningful contributions, demonstrating that UAFB and FSSA effectively extend the effective receptive field to almost all pixels. This capability allows FAT to capture multiscale and frequency-domain features globally, significantly improving SR performance.

In addition, we examined the influence of model parameters on performance by introducing a lightweight FAT-S model and a more powerful FAT-L model. In the FAT-S model, the channel number was set to 96, with five SRTBs, each containing two STLs and one MAT, and the W-MSA window size set to 4. For the FAT-L model, the channel number increased to 192, with eight SRTBs, each containing three STLs and one MAT, and the W-MSA window size set to 6. Table VIII presents the model parameters and corresponding quantitative results. As the model size increased, FAT demonstrated superior SR capabilities for remote sensing images.

E. Evaluation of the Loss Function

We conducted a series of comparative and ablation experiments on the combined loss function, with Table IX summarizing the ablation study results. The local extremum and variance losses significantly improved the perceptual quality and SSIM performance of the SR images only with a slight decrease in PSNR. By emphasizing the local data

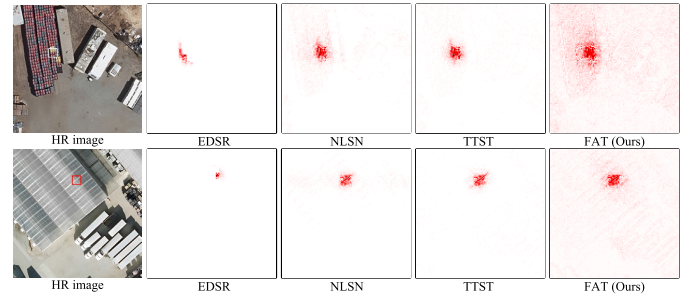


Fig. 14. Comparison of LAM results across different SR networks. The LAM attribution reflects the importance of each pixel in the input LR image for reconstructing texture regions highlighted by bounding boxes.

TABLE VII
ABLATION STUDY OF UAFB, SRTB, AND MAT

Method	UAFB	SRTB	MAT	PSNR	SSIM
Model-A	✗	✗	✗	28.932	0.7988
Model-B	✗	✓	✓	29.412	0.8131
Model-C	✓	✗	✓	29.264	0.8089
Model-D	✓	✓	✗	29.181	0.8058
FAT(Ours)	✓	✓	✓	29.460	0.8148

TABLE VIII
COMPARISON WITH DIFFERENT SIZES OF FAT, CALCULATED WITH A $3 \times 120 \times 120$ IMAGE INPUT

Method	#Param.	FLOPS	PSNR	SSIM
FAT-S	4.413M	69.060G	29.341	0.8105
FAT	10.192M	153.513G	29.460	0.8148
FAT-L	29.036M	418.782G	29.521	0.8180

distribution and high-frequency textures, the proposed loss function further enhances the realism of reconstructed images when combined with perceptual loss. Notably, the proposed loss function enabled FAT to achieve a remarkable 9.6% reduction in LPIPS with only a 0.007 dB drop in PSNR. In addition, the local extremum and variance losses can be applied to other models to mitigate inconsistencies in texture reconstruction across different regions, thereby improving the overall quality of the SR results.

To evaluate the impact of the proposed loss function on high-frequency texture reconstruction, we performed qualitative visual comparisons of the SR results. In addition, we visualized the Fourier spectrum of these results.

TABLE IX
COMPARISON STUDY OF LOSS FUNCTION

Method	L_1	$L_{\text{perceptual}}$	L_{local}	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
FAT	✓			29.460	0.8148	0.1672
	✓	✓		29.448	0.8122	0.1457
	✓		✓	29.453	0.8175	0.1512
	✓	✓	✓	29.444	0.8167	0.1278
HAT	✓			29.422	0.8123	0.1747
	✓	✓		29.404	0.8100	0.1531
	✓		✓	29.408	0.8146	0.1593
	✓	✓	✓	29.397	0.8137	0.1328

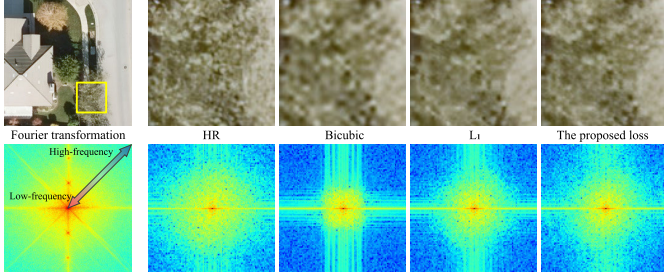


Fig. 15. Comparison between HR and bicubic images, as well as SR results driven by different loss functions and their corresponding Fourier transform outcomes.

TABLE X
EFFECTS OF THE WEIGHTING FACTOR λ_1 IN L_{LOCAL}

λ_1	0	0.1	1	10
PSNR $\uparrow$	29.450	29.449	29.444	29.329
SSIM $\uparrow$	0.8149	0.8162	0.8167	0.8091
LPIPS $\downarrow$	0.1369	0.1363	0.1278	0.1104

TABLE XI
EFFECTS OF THE WEIGHTING FACTOR λ_2 IN L_{LOCAL}

λ_2	0	0.1	1	10
PSNR $\uparrow$	29.449	29.450	29.444	29.407
SSIM $\uparrow$	0.8156	0.8162	0.8167	0.8100
LPIPS $\downarrow$	0.1305	0.1286	0.1278	0.1224

As illustrated in Fig. 15, compared with the low-resolution input images, the FAT[†] model driven by L_1 loss excels in reconstructing mid-frequency details. However, when driven by the proposed loss function, the intensity of high-frequency features in the SR results is significantly enhanced. Consequently, the Fourier spectrum of the reconstructed images more closely resembles that of the high-resolution ground truth.

We further examined the contributions of the internal coefficients in the proposed local extremum and variance loss functions to the SR results. As mentioned below, these losses consist of a weighted sum of the local extremum loss L_{ext} and the local variance loss L_{var} . By adjusting the weight λ_1 for L_{ext} and λ_2 for L_{var} , we quantified their contribution to SR performance. The results presented in Tables X and XI confirm our analysis. Increasing these weights leads to a decline in PSNR performance but markedly improves perceptual quality.

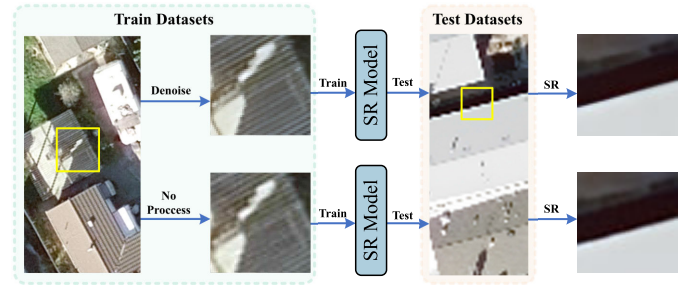


Fig. 16. Workflow and effects of denoising-based dataset augmentation.

TABLE XII
QUANTITATIVE RESULTS OF DATA AUGMENTATION STRATEGY

Augmentation	PSNR	SSIM	LPIPS
w/ Denoising	29.118	0.7993	0.1699
w/o Denoising	28.321	0.7788	0.2389

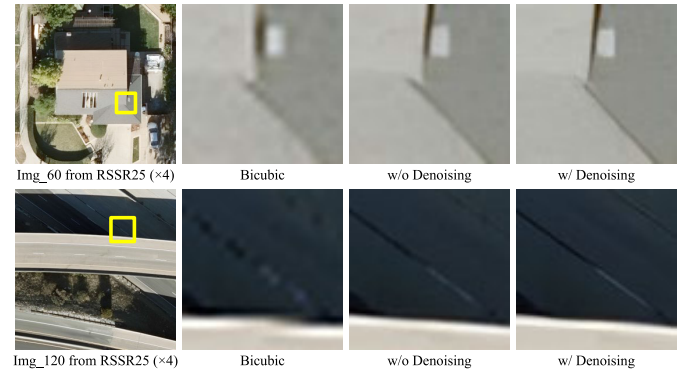


Fig. 17. Qualitative comparison of the impact of data augmentation strategy on SR results.

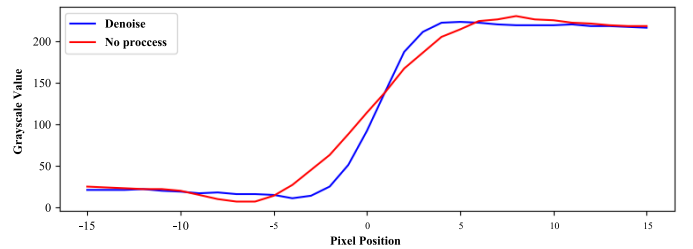


Fig. 18. Example comparison of the results of restoring the same edge by the model using different dataset preprocessing methods.

After balancing PSNR, SSIM, LPIPS, and the visual quality of SR results, we set $\lambda_1 = 1$ and $\lambda_2 = 1$.

F. Datasets Discussion

To evaluate the effectiveness of dataset augmentation based on denoising algorithms in improving SR performance, we conducted a comparative experiment. The experimental process is illustrated in Fig. 16. We applied a denoising version of SwinIR to enhance the dataset, reducing artifacts and noise. The proposed FAT model was trained on the enhanced DOTA and AID datasets and tested on the high-quality RSSR25 dataset. For comparison, we also trained the model on the

original, unprocessed DOTA and AID datasets and evaluated it on RSSR25.

The quantitative results in Table XII show that applying the dataset augmentation strategy led to a significant 0.798-dB improvement in PSNR, along with a substantial 29% enhancement in perceptual quality. These findings indicate that dataset augmentation effectively boosts SR performance and model generalization.

The qualitative comparison in Fig. 17 further supports this conclusion. Without augmentation, the SR results exhibit blurry edges and ringing artifacts during reconstruction. In contrast, SR images generated with dataset augmentation display sharper and more distinct contours, achieving superior reconstruction quality.

To further investigate the impact of denoising-based augmentation on SR performance, we conducted a data analysis by extracting pixel values from the SR output of a contour edge. As shown in Fig. 18, the presence of noise and artifacts in the unprocessed dataset results in SR images with slower pixel transitions, leading to excessive smoothing and ringing artifacts. In contrast, images generated by the model with dataset augmentation exhibit sharper and more realistic edges, leading to better visual quality. These findings indicate that noise introduced during dataset collection and preparation plays a crucial role in edge blurring and ringing artifacts in existing models. Given the scarcity of high-quality data, denoising algorithms can serve as an effective low-quality data augmentation strategy to enhance SR performance and model generalization.

V. CONCLUSION

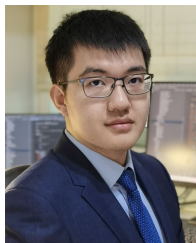
In this study, we propose FAT to enhance remote sensing image SR by focusing on high-frequency texture details and mitigating the model's bias toward different textures. To achieve this, we introduce key modules such as UAFB and FSSA, which efficiently extract frequency-domain features and model frequency-domain correlations, significantly improving SR performance. Furthermore, we analyze the limitations of existing loss functions and develop a local extremum and variance loss function, which guides the model to focus on local data distribution and high-frequency textures, achieving an exceptional balance between distortion and perceptual quality. In addition, we construct a high-quality remote sensing SR dataset, RSSR25, and explore the potential of denoising algorithms as an augmentation strategy to enhance model performance and generalization. Extensive experiments demonstrate the effectiveness of our proposed model in remote sensing image SR, consistently surpassing state-of-the-art methods in both visual quality and quantitative metrics. Similar to many previous studies, our research primarily relies on synthetic datasets. However, real-world remote sensing image degradation is influenced by multiple complex factors, and degradation models designed for natural images often overlook unique remote sensing degradations, such as stripe noise. In future work, we plan to develop a degradation model that better reflects the real-world degradation characteristics of remote sensing images, further enhancing the engineering applicability of FAT.

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